

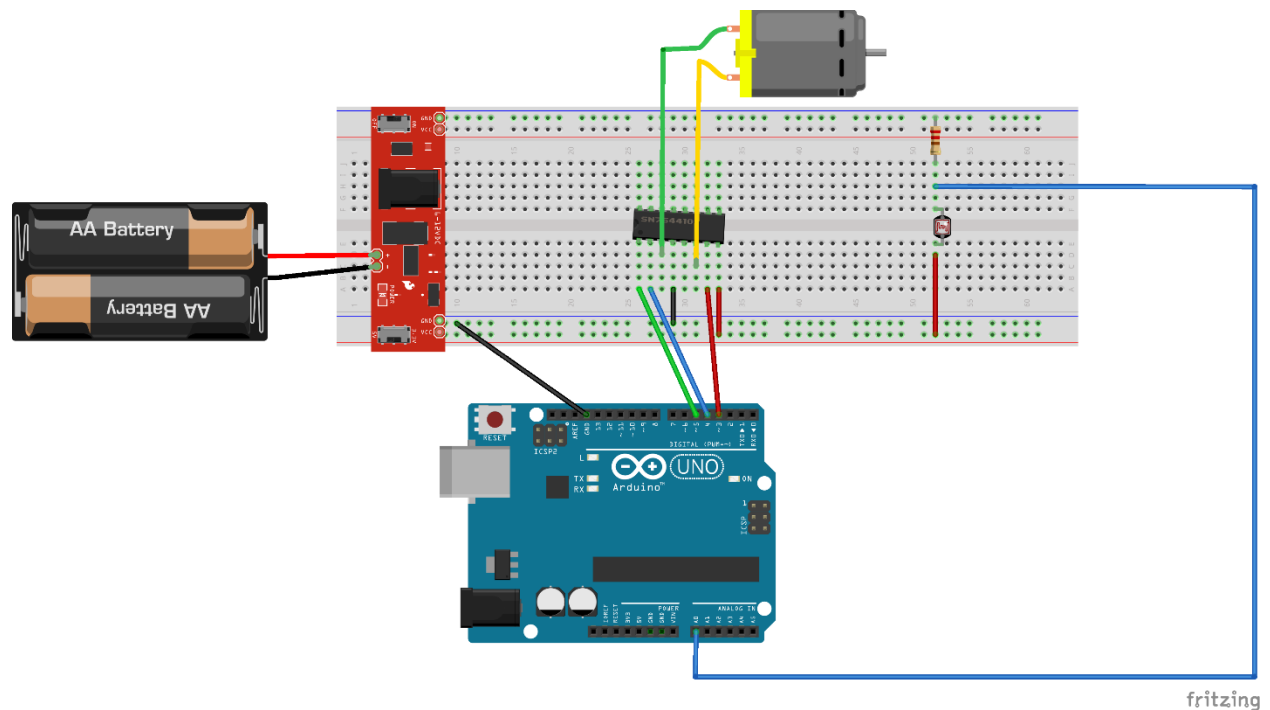
Motion Detector Motor:

I created a circuit in Arduino that would activate a DC motor when the photoresistor resistance value increases above a certain threshold then the Arduino microcontroller will send a signal to enable the motor and set a direction for the motor to spin (clockwise or counterclockwise). When the resistance decreases in the circuit, the MCU disables the DC motor and the motor stops. The resistance in a photoresistor decreases when light hits the photocell and increases as light is blocked from the photocell. Ergo, when a hand passes over the photocell, the motor activates as the hand's shadow blocks light which increases resistance.

Link to Performance Video:

<https://youtube.com/shorts/6mWikQEEY0g?feature=share>

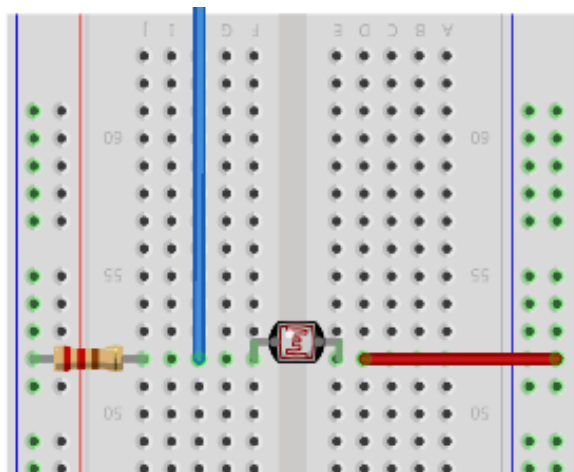
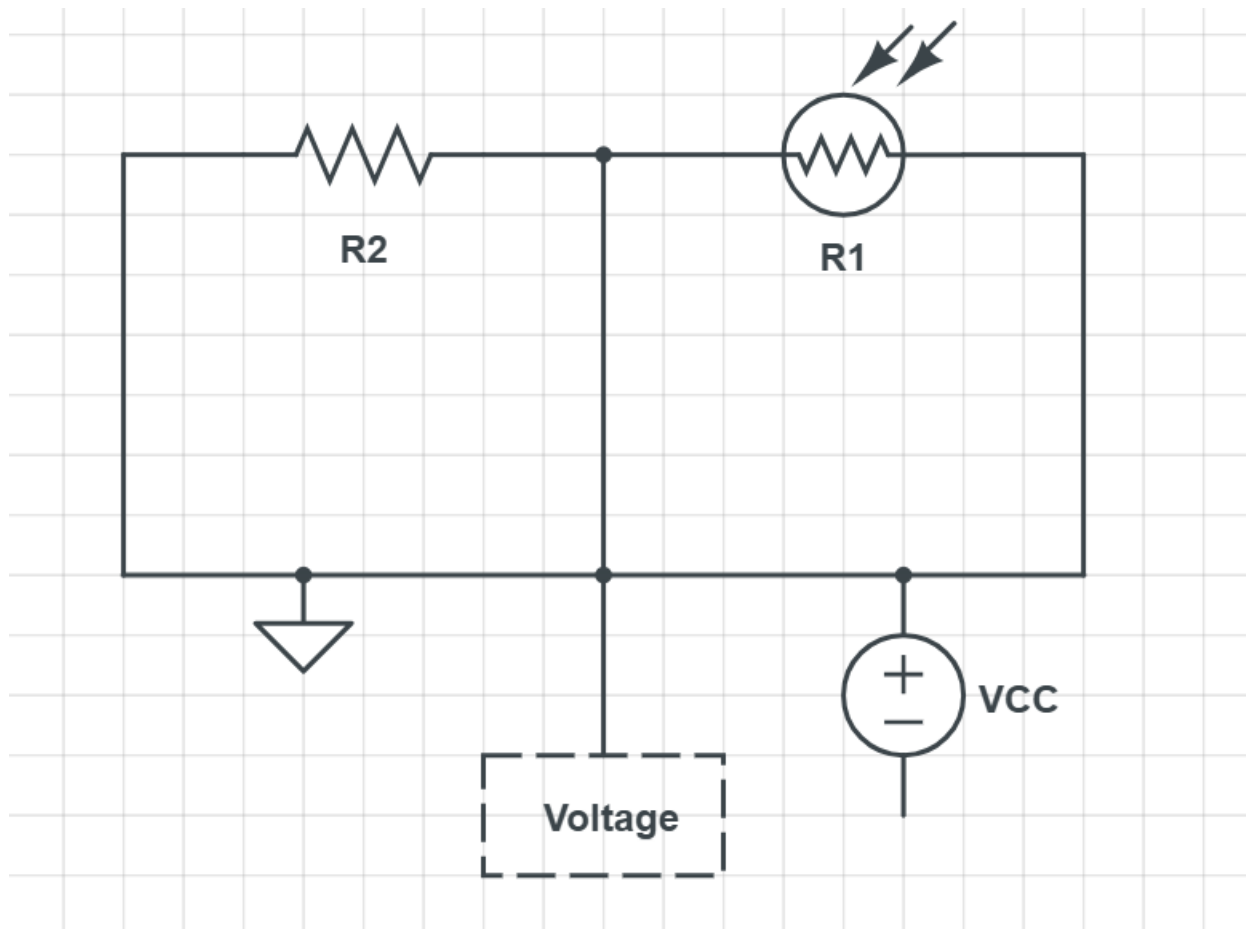
Schematic and Parts:



Arduino analog pins do not measure resistance, but voltage.

$$V = V_{CC} \cdot \frac{R_2}{R_1 + R_2}$$

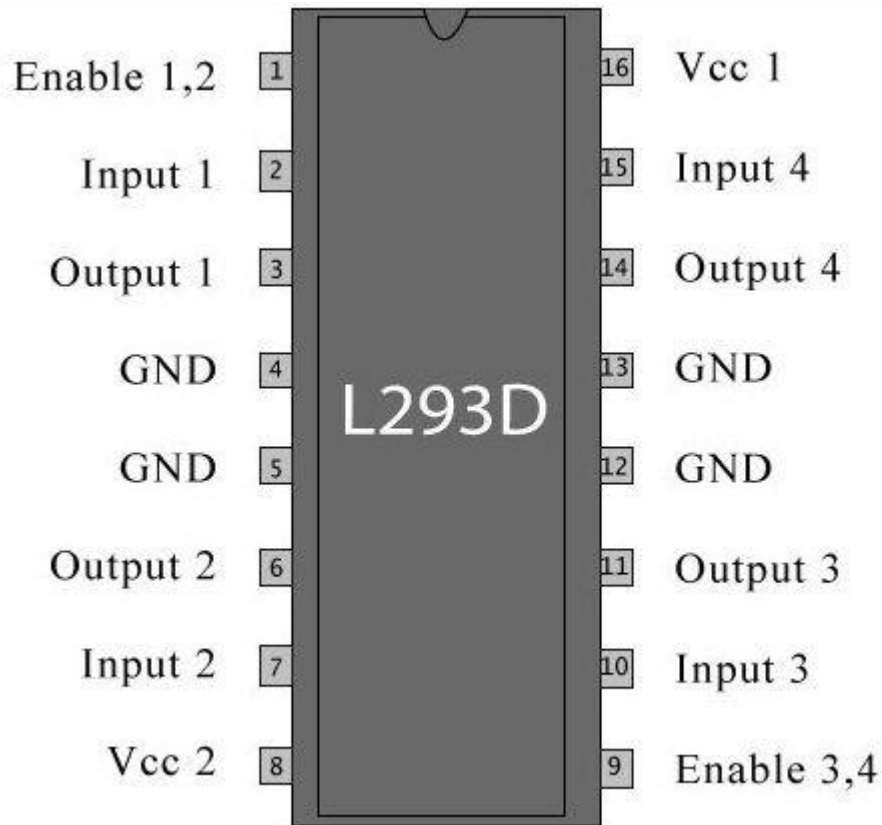
Where in this case R_1 is the photoresistor and R_2 is 220 Ohm resistor in series with the photoresistor in a voltage divider circuit. As light is blocked from R_1 via an obstacle, its resistance increases and the signal voltage decreases.



When the Analog voltage sensor reads as under 40, the micro controller writes the digital pin 5 (Output) as HIGH to enable the motor from the first pin of the L293D

Pin 3 is written as HIGH at the same time which goes to Input 2 on the on the L239D, which sends the motor spinning counterclockwise as the terminal of the motor is activated.

When the Analog pin reads above 40 again, pin 5 is rewritten as LOW and the motor is disabled.



This relatively simple circuit was built using:

- Arduino Uno R3
- Elegoo Mb-v2 power supply
- 9 volt battery
- TI L239d Dual H-bridge Motor Driver
- 220 Ω Resistor
- A photoresistor
- DC Motor

When the analog voltage sensor reads voltage as

Arduino Code:

```
#define ENABLE 5  
#define DIRA 3
```

```
#define DIRB 4
int lightPin = 0;
int i;

void setup() {
  //---set pin direction
  pinMode(ENABLE,OUTPUT);
  pinMode(DIRA,OUTPUT);
  pinMode(DIRB,OUTPUT);
  Serial.begin(9600);
}

void loop() {
  //---back and forth example
  int reading = analogRead(lightPin);
  Serial.println(reading);
  if (reading <40){
    digitalWrite(ENABLE,HIGH); // enable on
    digitalWrite(DIRA,HIGH); //one way
    digitalWrite(DIRB,LOW);
  }
  else{
    digitalWrite(ENABLE,LOW); // disable
    // digitalWrite(DIRA,HIGH); //one way
    // digitalWrite(DIRB,LOW);
  }
}
```